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June 14, 1965
SUPERSEDING
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FEDERAL SPECIFICATION

CLEANING COMPOUND, SOLVENT-DETERGENT (FOR CLEANING COILS AND FILTERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The cleaning compound covered by this specification is for the external use in the cleaning of coils and filters of heating and air conditioning systems (see 6.1).

1.2 Classification. The cleaning compound covered by this specification shall be a liquid and of one grade only.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-A-250/2—Aluminum Alloy 3003, Plate and Sheet.

TT-E-489—Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).

TT-L-58—Lacquer, Spraying, Clear and Pigmented (General Use).

TT-P-636—Primer Coating, Alkyd, Wood and Ferrous Metal.

PPP-D-729—Drums: Metal, 55-gallon (For Shipment of Noncorrosive Material).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials: Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are

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available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–D–16791—Detergent, Liquid, Nonionic (Type I).

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The manufacturer is permitted wide latitude in the selection of raw materials providing the cleaning compound conforms to the requirements of this specification.

3.2 Stability in use. When tested as specified in 4.5.1, the compound shall not develop precipitates or sediments.

3.3 Miscibility with water. The compound shall be completely miscible with distilled water forming a clear solution when tested as specified in 4.5.2.

3.4 Flash point. The flash point of the liquid shall not be below 212° F. or below the boiling point of the concentrate as specified in 4.5.3.

3.5 pH value. The pH value of the concentrate shall be not less than 9.0 nor more than 9.5 at 25° C. as specified in 4.5.4.

3.6 Nonvolatile content. The residue on evaporation shall be no less than 18 percent, as specified in 4.5.5.

3.7 Nature of residue. The residue on evaporation shall be a heavy fluid of nondrying property as specified in 4.5.6.

3.8 Corrosion to steel. The solution of the compound shall not cause rusting or corrosion of mild steel as specified in 4.5.7.

3.9 Corrosion to aluminum. The cleaning compound shall not cause discoloration of aluminum as specified (see 4.5.8).

3.10 Brightening of copper. A use solution of the compound shall brighten a tarnished (remove tarnish) copper surface as specified in 4.5.9.

3.11 Cleaning properties. A diluted solution of water shall be equal in cleaning ability to a similar dilution of a comparison formula as described in 4.5.10.

3.12 Deleterious effect on painted surface. The compound shall not cause any deleterious effect when tested as described in 4.5.11.

3.13 Toxicity characteristics. The cleaning compound shall have no adverse effect on the health of personnel when used for its intended purpose.

3.14 Workmanship. The cleaning compound shall be manufactured in accordance with the best commercial practice to produce a high quality material suitable for the intended purpose.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or pur-

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chase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance.

4.2.1 *Inspection lot.* For purposes of sampling, a lot shall consist of one batch of the cleaning compound presented for inspection at one time.

4.2.2 *Sampling for examination of filled containers.* A random sample of filled containers shall be selected from each lot of cleaning compound in accordance with MIL–STD–105 at inspection level 1 and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking and other requirements not involving tests.

4.2.3 *Sampling for tests.* From each inspection lot, the Government inspector shall take two containers at random. From each of the two containers one quart of the compound shall be taken and placed in separate clean, dry, glass or polyethlene containers, sealed, marked and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.3 Inspection.

4.3.1 *Inspection of filled containers.* Each sample filled container taken in accordance with 4.2.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample-filled container shall also be gauged to determine the amount of the contents. Any container in the sample having one or more defects, or being under the required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination provided that the contractor has removed or repaired all nonconforming containers.

4.4 Lot examination tests. The sample specimens taken in accordance with 4.2.3 shall be subjected separately to the tests specified in 4.5. If either specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for examination tests provided the contractor has removed or reworked all nonconforming products.

4.5 Test procedures.

4.5.1 *Stability in use.* Place 50 cc. of the concentrate in one beaker and place 50 cc. diluted solution (one part plus one part of distilled water), into another beaker, let stand uncovered at room temperature and observe for turbidity or sediment in either one of the two solutions after three days. Turbidity or sediment in either one of the two solutions is cause for failure (see 3.2).

4.5.2 *Miscibility with water.* Dilute one part of the compound with nine parts of distilled water and after five minutes observe for separation, turbidity or sediment (see 3.3).

4.5.3 *Flash point.* Flash point of the concentrate shall be determined in accordance with Fed. Test Method Std. No. 141, method 4294 (see 3.4).

4.5.4 *pH value.* The pH value of the concentrate shall be determined by means of a Beckman pH meter or its equivalent at 25° C. (see 3.5).

4.5.5 *Nonvolatile content.* A 5.000 g. sample of the concentrate shall be weighed

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into a 100 mm. (4-inch) diameter watchglass and dried to a constant weight in a forced draft oven maintained at 100° C. After the evaporation period, let cool in a desiccator before weighing. The loss of volatiles shall be no more than 82 percent (see 3.6). Continue test with 4.5.6.

4.5.6 *Nature of residue.* The residue on the watchglass from test 4.5.5 shall be a viscuous fluid capable of flowing at room temperature. There shall be no noticeable nonhomogeneity of the residue, such as separation or sediments. Observe for same properties of the residue after placing watchglass under a bell jar after 72 hours.

4.5.7 *Corrosion to steel.* Dilute one part of the concentrate with nine parts of distilled water. Place 50 cc. of this solution in a 4 oz. bottle with a piece of 1½" by 3" polished S.A.E., 1020 cold-rolled steel. Cap bottle and shake vigorously for one minute. (Let stand undisturbed for one week and observe for rusting on the steel above the solution, at the solution line (interphase), and below the solution (see 3.8).

4.5.8 *Corrosion to aluminum.* Clean a piece of 3" by ½" aluminum strip (QQ–A–250/2, H18, 0.025 inch mill finish) with a clean cloth soaked in methyl ethyl ketone. Place cleaned aluminum strip in 50 ml. stoppered graduate with 8 millimeter of the compound (undiluted). Stopper graduate and without agitation, let stand at room temperature for 18-24 hours. After this period, rinse test aluminum strip in cold running water and wipe dry. Observe for signs of etching or discoloration in the solution phase, interphase and vapor phase of the aluminum test specimen.

4.5.9 *Brightening effects on copper.* Place a piece of copper tube, or plate which has been left in atmosphere and has been corroded or tarnished with a greenish tint, into a solution of the compound diluted one to one with distilled water. Let stand for one hour, and rinse the copper specimen in running tap water, let dry and observe for brightening effect by comparison with an untreated copper specimen (see 3.10).

4.5.10 *Cleaning properties.* Clean one-half section of a refrigeration or air conditioning air flow coil which has been in use daily for 90 days or more with a solution of the compound (one part concentrate with two parts of distilled water) and clean the other half section with a similarly diluted solution of a comparison formula as follows:

Sodium metasilicate pentahydrate	5 percent
Detergent, nonionic*	10 percent
Deionized water	85 percent

Note: This comparison formula does not fulfill the other requirements of this specification.

*Conforming to MIL–D–16791, type I.

Use appropriate pressurized sprayer to spray solution on coils, let stand for 15 minutes and flush with water. The cleaning property and rinsing characteristics of the compound shall be equal to or better than the comparison formula (see 3.11).

4.5.11 *Deleterious action on painted surface.* Use the same solutions as described in 4.5.9 and spray on painted panels prepared as follows:

Two clean aluminum alloy panels conforming to QQ–A–250/2 H18, 0.025 inch mill finish, shall be finished as shown in table I. The coat of zinc chromate primer shall be air-dried for two hours and then baked for two hours at 250° F. before application of the finish coats, and the lacquer shall be air-dried for one hour between coats. The finished panels shall be air-dried for two hours, then baked for 18 hours at 66° C. (150° F.), and allowed to cool at room temperature.

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TABLE I. *Preparation of panels for test*

Panel	First coat	Finish
1	Zinc chromate primer Fed. Spec. TT–P–636	One coat enamel Fed. Spec. TT–E–489 class A.
2	Zinc chromate primer Fed. Spec. TT–P–636	Two coats lacquer Fed. Spec. TT–L–58 type II

The panels are placed at 45° angle so that excessive spray will run down the panels. After a period of one hour, rinse each panel with distilled water and blow dry with compressed air. The panels shall then be examined for discoloration, softening or other defects.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of levels of packing shall be in accordance with Fed. Std. No. 102.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* The product shall be filled in 55 gallon capacity metal drums. The metal drums shall conform to PPP–D–729, type I. Cap seals shall be provided. Closure shall be in accordance with the requirements of the drum specification.

5.1.2 *Level B.* The product shall be filled in 55 gallon capacity metal drums. The metal drums shall conform to PPP–D–729, type II, except that unless otherwise specified, commercial exterior coating will be acceptable. Closure shall be in accordance with the requirements of the drum specification.

5.1.3 *Level C.* The product shall be filled in containers complying with the rules and regulations applicable to the mode of transportation to insure carrier acceptance and safe delivery at lowest transportation cost to destination.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL–STD–129.

5.2.3 *Special markings.* Special markings and warnings shall be shown on the shipping container as required by the Interstate Commerce Commission and such other statutory regulations as are applicable. Such markings and warnings shall be affixed to the container in a conspicuous location. In addition, each container shall be labeled with use instructions.

6. NOTES

6.1 Intended use. The cleaning compound covered by this specification is for use as a cleaning solvent for the cleaning of coils and filters in heating and air conditioning systems, fans, refrigeration systems and may also be used as a nonflammable filter adhesive which will improve filter action and ease subsequent cleaning.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Applicable levels of packing (5.1) and special markings required (5.2).

6.3 Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Cleaning compounds, not otherwise indexed by name, liquid.

Carload minimum weight 36,000 pounds.

Motor:

Cleaning compound, not otherwise indexed, liquid.

Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—SH

Air Force—69

Review Interest:

Army—GL, MO, MD

Navy—SH, WP, YD

Air Force—69

User Interest:

Navy—MC

Copies of this specification may be purchased for 5 cents each.

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